

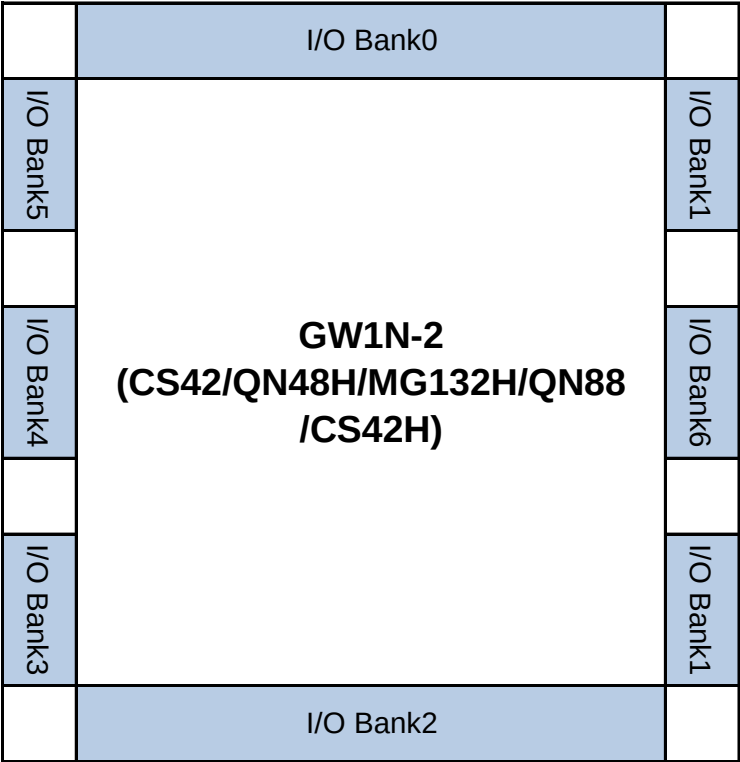
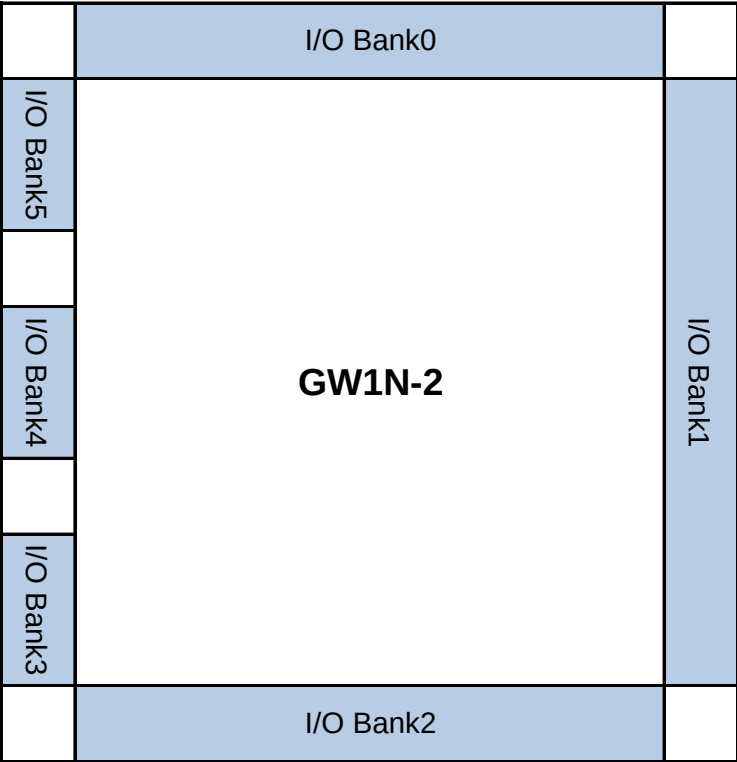
Date	Version	Description
11/23/2020	1.0E	Initial version published. CS42, MG132X, LQ100X, and LQ144X packages supported.
01/20/2021	1.1E	The info. of QN48 and QN48M packages added.
02/24/2021	1.2E	The info. of MG132 package added.
05/14/2021	1.3E	MG132 package is renamed MG132H, QN48M package is renamed QN48H. MG132, LQ100, and LQ144 packages added.
06/07/2021	1.3.1E	The descriptions of configuration functions for the pin IOT14A/IOT14B in MG132, LQ100, LQ144 packages added.
07/21/2021	1.3.2E	The info. of MG121 and MG121X packages added.
10/29/2021	1.4E	Pin definitions updated. The info. of MG49 package added.
12/03/2021	1.4.1E	The package info. of I ² C configuration mode updated.
01/20/2022	1.5E	The info. of QN32X, QN88, and CS42H packages added.
03/25/2022	1.5.1E	Pin definitions updated. The info. of CS42H package added.
06/15/2022	1.6E	The info. of QN32 and CS100H packages added. The pin and power info. of QN32X package updated.
07/08/2022	1.7E	The info. of LQ144F package added. QN88 package separated into UV and LV versions.
08/19/2022	1.7.1E	The pin info. of QN32X and QN32 packages in Pin List updated.
09/08/2022	1.7.2E	The I/O of Pin RECONFIG_N and the note in Pin Definitions sheet updated. The core voltage of QN88 package (UV version) in Power sheet updated.
11/11/2022	1.7.3E	The description of pin CLKHOLD_N in Pin Definitions sheet updated. The recommended operating conditions of QN32 package in Power sheet updated.
05/04/2023	1.7.4E	Power sheet updated. Pin definitions sheet updated. LQ144F (UV Version) package removed.
06/30/2023	1.7.5E	The descriptions of pin definitions and info. for MODE0, MODE1, MODE2 pins optimized.
11/30/2023	1.7.6E	The I/O descriptions of Ready and Done pins in Pin Definitions sheet optimized. The Min. value of I/O Bank voltage for CS32H, LQ144F, CS100H packages updated.
12/07/2023	1.7.7E	The descriptions of VCCD and VCCIOD pins in Pin Definitions sheet updated.
02/22/2024	1.7.8E	The note of VCCD and VCCIOD pins in Power sheet added.
06/28/2024	1.7.9E	The Min. value of VCCX voltage for LQ100X/LQ100 packages in Power sheet updated.
11/15/2024	1.8E	The description of EPAD pin in Pin Definitions sheet added.

Date	Version	Description
11/28/2025	1.8.1E	The EOL note of LQ144F package added.

Pin Name	I/O	Description
User I/O		
IO[End][Row/Column Number][A/B]	I/O	[End] indicates the pin location, including L(left), R(right), B(bottom), and T(top).
		[Row/Column Number] indicates the pin row/column number. If [End] is T(top) or B(bottom), the pin indicates the column number of the corresponding CFU. If [End] is L(left) or R(right), the pin indicates the row number of the corresponding CFU.
		[A/B] indicates differential signal pair information.
[pin]_[End][Row/Column Number][A/B]	I/O	[pin] indicates pin name. The info. of [End], [Row/Column Number], [A/B] is as above.
		Shorted IOs with pins are ranked by number. The first IO uses the pin name. Other shorted IOs use the pin name and the location information removed the character "IO". For example, IOL6A/IOL8A are shorted, its pin name is E6, then IOL6A corresponds to E6, IOL8A corresponds to E6_L8A.
Multi-Function Pins		
IO[End][Row/Column Number][A/B]/MMM		/MMM represents one or more of the other functions in addition to being general purpose user I/O. When these functions are not in use, these pins can be used as user I/O.
D0	I/O	Data port D0 in CPU mode
D1	I/O	Data port D1 in CPU mode
D2	I/O	Data port D2 in CPU mode
D3	I/O	Data port D3 in CPU mode
D4	I/O	Data port D4 in CPU mode
D5	I/O	Data port D5 in CPU mode
D6	I/O	Data port D6 in CPU mode
D7	I/O	Data port D7 in CPU mode
WE_N	I	Select data input/output of D[7:0] in CPU mode.0: Write;1: Read.
DOUT	O	Data output in SERIAL mode
DIN	I, internal weak pull-up	Data input in SERIAL mode
TMS	I, internal weak pull-up	Serial mode input in JTAG mode
TCK	I	Serial clock input in JTAG mode
TDO	O	Serial data output in JTAG mode
TDI	I, internal weak pull-up	Serial data input in JTAG mode
JTAGSEL_N	I, internal weak pull-up	Reconfigure JTAG download function signal
RECONFIG_N	I, internal weak pull-up	Global reset GowinCONFIG logic signal, active low
FASTRD_N	I	Access SPI FLASH to select signal. Low, Fast Read mode; High, Read mode.
DONE ^[1]	O, internal weak pull-up	High, the programming configuration has been completed successfully; Low, the programming configuration has not been completed or failed.
	I, internal weak pull-up	When the DONE signal is low, delay the chip to activate. Activate the chip until the DONE signal is high.

Pin Name	I/O	Description
READY ^[1]	I/O, internal weak pull-up	High, the device can be programmed and configured currently; Low, the device cannot be programmed and configured currently.
MI	I	MI in MSPI mode
MO	O	MO in MSPI mode
MCS_N	O	Enable signal MCS_N in MSPI mode, active-low
MCLK	O	Clock output MCLK in MSPI mode, with default frequency of 2.5Mhz
SCLK	I	Clock input in SSPI, SERIAL, and CPU modes
SO	O	SO in SSPI mode
SI	I/O	SI in SSPI mode
SSPI_CS_N	I/O	Enable signal SSPI_CS_N in SSPI mode, active-low, and internal weak pull-up
SDA	I/O	I2C serial data line: When I2C is supported in GowinCONFIG mode, external pull-up is required.
SCL	I	I2C serial clock line: When I2C is supported in GowinCONFIG mode, external pull-up is required.
CLKHOLD_N	I, internal weak pull-up	Active-high in SSPI mode; Active-low in CPU mode.
GCLKC_[x]	I	Differential input pin of GCLKT_[x], C(Comp), [x] is global clock number ^[2]
GCLKT_[x]	I	Global clock input pin, T(True), [x] is global clock number
LPLL_C_fb/RPLL_C_fb	I	Left/Right PLL feedback input pin, C(Comp)
LPLL_T_fb/RPLL_T_fb	I	Left/Right PLL feedback input pin, T(True)
LPLL_C_in/RPLL_C_in	I	Left/Right PLL clock input pin, C(Comp)
LPLL_T_in/RPLL_T_in	I	Left/Right PLL clock input pin, T(True)
MODE2	I, internal weak pull-up	GowinCONFIG modes selection pin; if this pin is marked as "VCCIO", it's internally powered; if this pin is marked as "GND", it's internally grounded.
MODE1	I, internal weak pull-up	GowinCONFIG modes selection pin; if this pin is marked as "VCCIO", it's internally powered; if this pin is marked as "GND", it's internally grounded.
MODE0	I, internal weak pull-up	GowinCONFIG modes selection pin; if this pin is marked as "VCCIO", it's internally powered; if this pin is marked as "GND", it's internally grounded.
Other Pins		
CKP	DIO ^[3]	The clock channel input pin of MIPI_DPHY_RX
CKN	DIO ^[3]	The clock channel differential pin of MIPI_DPHY_RX
RX0P	DIO ^[3]	The data channel 0 input pin of MIPI_DPHY_RX
RX0N	DIO ^[3]	The data channel 0 differential pin of MIPI_DPHY_RX
RX1P	DIO ^[3]	The data channel 1 input pin of MIPI_DPHY_RX

Pin Name	I/O	Description
RX1N	DIO ^[3]	The data channel 1 differential pin of MIPI_DPHY_RX
RX2P	DIO ^[3]	The data channel 2 input pin of MIPI_DPHY_RX
RX2N	DIO ^[3]	The data channel 2 differential pin of MIPI_DPHY_RX
RX3P	DIO ^[3]	The data channel 3 input pin of MIPI_DPHY_RX
RX3N	DIO ^[3]	The data channel 3 differential pin of MIPI_DPHY_RX
NC	NA	Reserved
VSS	NA	Ground
VCC	NA	Power supply pin of core voltage
VCCD	NA	Power supply pin of MIPI
VCCIOD	NA	Power supply pin of MIPI
VCCIO#	NA	Power supply pin of I/O voltage for I/O BANK#
VCCX	NA	Power supply pin of auxiliary voltage
EPAD	NA	Exposed pad. Connect to ground.
Note! ^[1] The default state of READY/DONE is open-drain output, internal weak pull-up. DONE outputs 0 during configuration. ^[2] When the input is single-ended, GCLKC_[x] pin is not a global clock pin. ^[3] DIO indicates a dedicated pin.		



Note!

- [1] Each Bank has independent reference voltage (VREF).
- [2] You can select to use IOB internal VREF (equals to 0.5 * VCCIO).
- [3] You can also select to use external VREF input (use any I/O pins as external VREF input).

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Note!

[1] UV Version

[2] LV Version

[3] The package is in the EOL stage.

Pin Name	Function	BANK	Configuration Function	Differential Pair	LVDS	X16	CS42[2]	MG132X[1]	MG132X[2]	LQ100X[1]	LQ100X[2]	LQ144X[1]	LQ144X[2]	QN48	QN48H[2]	MG132H	MG132[1]	MG132[2]	LQ100[1]	LQ100[2]	LQ144[1]	LQ144[2]	MG121X[1]	MG121X[2]	MG121[1]	MG121[2]	MG49	QN32X[1]	QN32X[2]	QN88[1]	QN88[2]	CS42H[2]	QN32[1]	QN32[2]	CS100H[2]	LQ144F[2][3]				
BANK5 True LVDS Pair																																								
IOL4A/LPLL_T_fb	I/O	5	LPLL_T_fb	True_of_IOL4B	True	x16	B5	B1	B1			1	1	15	16	B1	B1	B1			1	1	C9	C9	C9	C9					3	3					C10	1		
IOL4B/LPLL_C_fb	I/O	5	LPLL_C_fb	Comp_of_IOL4A	True	none	B6	B2	B2			2	2	14	17	B2	B2	B2			2	2	E8	E8	E8	E8					5	5					C9	2		
IOL6A/GCLKT_7	I/O	5	GCLKT_7	True_of_IOL6B	True	x16	D5	C2	C2	3	3	5	5	18	18	C2	C2	C2	3	3	5	5	C10	C10	C10	C10					6	6					D10	5		
IOL6B/GCLKC_7	I/O	5	GCLKC_7	Comp_of_IOL6A	True	none	D6	D1	D1	4	4	6	6	19	19	D1	D1	D1	4	4	6	6	C11	C11	C11	C11					7	7					D9	6		
IOL8A	I/O	5		True_of_IOL8B	True	x16	C5	E3	E3	9	9	11	11			E3	E3	E3	9	9	11	11	D10	D10	D10	D10					8	8					D9_L8A	11		
IOL8B	I/O	5		Comp_of_IOL8A	True	none	C6	F2	F2	10	10	12	12			F2	F2	F2	10	10	12	12	D11	D11	D11	D11					9	9					E10	12		
BANK4 True LVDS Pair																																								
IOL11A	I/O	4		True_of_IOL11B	True	x16		H1	H1			15	15	20	20	H1	H1	H1			15	15	E10	E10	E10	E10					10	10					E10_L11A	15		
IOL11B	I/O	4		Comp_of_IOL11A	True	none		H3	H3			17	17	21	21	H3	H3	H3			17	17	E11	E11	E11	E11					11	11					D10_L11B	17		
IOL13A	I/O	4		True_of_IOL13B	True	x16	E5	J1	J1	14	14	21	21			J1	J1	J1	14	14	21	21	G11	G11	G11	G11	D1				14	14					G9	21		
IOL13B	I/O	4		Comp_of_IOL13A	True	none	E6	J2	J2	15	15	22	22			J2	J2	J2	15	15	22	22	G10	G10	G10	G10	D2				13	13					G10	22		
BANK3 True LVDS Pair																																								
IOL15A	I/O	3		True_of_IOL15B	True	x16	F5	J3	J3	18	18	25	25	23	23	J3	J3	J3	18	18	25	25	G9	G9	G9	G9					16	16					G10_L15A	25		
IOL15B	I/O	3		Comp_of_IOL15A	True	none	F6	K2	K2	19	19	26	26	22	22	K2	K2	K2	19	19	26	26	G8	G8	G8	G8					15	15					E9	26		
IOL17A	I/O	3		True_of_IOL17B	True	x16	G5	M1	M1	24	24	32	32	27	27	M1	M1	M1	24	24	32	32	K11	K11	K11	K11	F1				19	19					E9_L17A	34		
IOL17B	I/O	3		Comp_of_IOL17A	True	none	G6	M2	M2	25	25	33	33	24	24	M2	M2	M2	25	25	33	33	J10	J10	J10	J10	F2				18	18					G9_L17B	31		
BANK2 True LVDS Pair																																								
IOB11A	I/O	2		True_of_IOB11B	True	x16		P8	P8	40	40	57	57			P8	P8	P8	40	40	57	57	K8	K8	K8	K8											K5	57		
IOB11B	I/O	2		Comp_of_IOB11A	True	none		M8	M8	41	41	58	58			M8	M8	M8	41	41	58	58	L8	L8	L8	L8											J5	58		
IOB13A	I/O	2		True_of_IOB13B	True	x16		M9	M9			61	61	33	33	M9	M9	M9			61	61	H9	H9	H9	H9	G5				38	38					H5	61		
IOB13B	I/O	2		Comp_of_IOB13A	True	none		N10	N10			62	62	32	32	N10	N10	N10			62	62	J9	J9	J9	J9	F5				37	37					G5	62		
IOB15A	I/O	2		True_of_IOB15B	True	x16		M10	M10			65	65			M10	M10	M10			65	65	K9	K9	K9	K9											K4	65		
IOB15B	I/O	2		Comp_of_IOB15A	True	none		P11	P11			67	67			P11	P11	P11			67	67	L9	L9	L9	L9											J4	67		
IOB17A	I/O	2		True_of_IOB17B	True	x16		M11	M11							M11	M11	M11					K10	K10	K10	K10											H4			
IOB17B	I/O	2		Comp_of_IOB17A	True	none		P12	P12							P12	P12	P12					L10	L10	L10	L10											G4			
IOB2A	I/O	2		True_of_IOB2B	True	x16		P2	P2			38	38			P2	P2	P2			38	38	L3	L3	L3	L3					26	26					K10	38		
IOB2B	I/O	2		Comp_of_IOB2A	True	none		N2	N2			39	39			N2	N2	N2			39	39	K3	K3	K3	K3					25	25					J10	39		
IOB4A	I/O	2		True_of_IOB4B	True	x16		N3	N3	29	29	42	42			N3	N3	N3	29	29	42	42	J5	J5	J5	J5	G1											42		
IOB4B	I/O	2		Comp_of_IOB4A	True	none		P4	P4	30	30	43	43			P4	P4	P4	30	30	43	43	J4	J4	J4	J4	G2										H8	43		
IOB6A	I/O	2		True_of_IOB6B	True	x16		N5	N5			47	47			N5	N5	N5			47	47	L5	L5	L5	L5					32	32					K8	47		
IOB6B	I/O	2		Comp_of_IOB6A	True	none		M5	M5			48	48			M5	M5	M5			48	48	K5	K5	K5	K5					31	31					J8	48		
IOB8A	I/O	2		True_of_IOB8B	True	x16		P7	P7	36	36	52	52			P7	P7	P7	36	36	52	52									35	35					H7	52		
IOB8B	I/O	2		Comp_of_IOB8A	True	none		N7	N7	37	37	54	54			N7	N7	N7	37	37	54	54									36	36					H6	54		
BANK1 True LVDS Pair																																								
IOR11A/GCLKT_2	I/O	1	GCLKT_2	True_of_IOR11B	True	none		G13	G13	63	63	92	92	41		K13	G13	G13	63	63	92	92	F2	F2	F2	F2	E7	21	21							E3	21	21	G3	94
IOR11B/GCLKC_2	I/O	1	GCLKC_2	Comp_of_IOR11A	True	none		H12	H12	62	62	91	91	40		K14	H12	H12	62	62	91	91	F1	F1	F1	F1	E6	20	20						F3	20	20	F3	93	
IOR13A	I/O	1		True_of_IOR13B	True	none		G12	G12	61	61	89	89	38		L14	G12	G12	61	61	89	89	G1	G1	G1	G1												H1	89	
IOR13B	I/O	1		Comp_of_IOR13A	True	none		G14	G14			87	87	39		M13	G14	G14			87	87	G2	G2	G2	G2												H2	87	
IOR15A	I/O	1		True_of_IOR15B	True	none		J13	J13			84	84			M12	J13	J13			84	84	H1	H1	H1	H1	G6											J1	84	
IOR15B	I/O	1		Comp_of_IOR15A	True	none		K12	K12			83	83			M14	K12	K12			83	83	H2	H2	H2	H2	G7											J2	83	
IOR17A	I/O	1		True_of_IOR17B	True	none		L14	L14			78	78			N13	L14	L14			78	78	J2	J2	J2	J2												K1	78	
IOR17B	I/O	1		Comp_of_IOR17A	True	none		M13	M13			77	77			N14	M13	M13			77	77	J3	J3	J3	J3												K2	77	
IOR1A	I/O	1		True_of_IOR1B	True	none		B14	B14			107	107	47		B14	B14	B14			107	107	C2	C2	C2	C2												A1	107	
IOR1B	I/O	1		Comp_of_IOR1A	True	none		C13	C13			106	106	46		C13	C13	C13			106	106	D3	D3	D3	D3												A2	106	
IOR3A/D2	I/O	1	D2	True_of_IOR3B	True	none		C14	C14	71	71	100	100	45		C14	C14	C14	71	71	100	100	D2	D2	D2	D2	E5												B1	103
IOR3B/D3	I/O	1	D3	Comp_of_IOR3A	True	none		D12	D12	70	70	99	99	44		D12	D12	D12	70	70	99	99	D1	D1	D1	D1	E4												B2	
IOR5A/MID7	I/O	1	MID7	True_of_IOR5B	True	none		E12	E12	67	67	96	96	43		E12	E12	E12	67	67	96	96	E2	E2	E2	E2												C3	98	
IOR5B/MID6	I/O	1	MID6	Comp_of_IOR5A	True	none		E14	E14	66	66	95	95	42		E14	E14	E14	66	66	95	95	E1	E1	E1	E1					62	62						D4	97	
BANK0 True LVDS Pair																																								
IoT11A/GCLKT_0	I/O	0	GCLKT_0	True_of_IoT11B	True	x16		A7	A7	88	88	128	128	6	8	A7	A7	A7	88	88	128	128	B6	B6	B6	B6					77	77						D7	128	
IoT11B/GCLKC_0	I/O	0	GCLKC_0	Comp_of_IoT11A	True	none		B7	B7	87	87	127	127	7	9	B7	B7	B7	87	87	127	127	A6	A6	A6	A6					78	78							B6	127
IoT13A	I/O	0		True_of_IoT13B	True	x16	B3	A10	A10	84	84	122	122			A10	A10	A10	84	84	122	122	A4	A4	A4	A4	A5													

Note!

VCCX should be greater than or equal to the Max. VCCIO.

^[1] When MIPI HARD Core is not used, users can either leave VCCD and VCCIOD floating or connect them to 1.2V power.

^[2] The package is in the EOL stage.

Recommended Operating Conditions of CS42 Package in GW1N-2

Name	Description	Min.	Max.
VCC	Core voltage	1.14V	1.26V
VCCIO0/VCCIO2	VCCIO0 and VCCIO2 are internally short-circuited.	1.14V	3.6V
VCCIO3/VCCIO4/VCCIO5	VCCIO3, VCCIO4 and VCCIO5 are internally short-circuited.	1.14V	3.6V
VCCD/VCCIO1/VCCIOD ^[1]	VCCD, VCCIO1, and VCCIOD are internally short-circuited.	1.14V	1.26V
VCCX	Auxiliary voltage	1.71V	3.6V

Recommended Operating Conditions of MG132X/MG132/MG121X/MG121 Package in GW1N-2 (LV Version)

Name	Description	Min.	Max.
VCC	Core voltage	1.14V	1.26V
VCCIO0, VCCIO2, VCCIO3, VCCIO4, VCCIO5	I/O Bank voltage	1.14V	3.6V
VCCX/VCCIO1	Auxiliary voltage VCCX and VCCIO1 are internally short-circuited.	1.71V	3.6V

Recommended Operating Conditions of MG132X/MG132/MG121X/MG121 Package in GW1N-2 (UV Version)

Name	Description	Min.	Max.
VCCX/VCC	Core voltage VCC and auxiliary voltage VCCX are internally short-circuited.	1.71V	3.6V
VCCIO0, VCCIO1, VCCIO2, VCCIO3, VCCIO4, VCCIO5	I/O Bank voltage	1.14V	3.6V

Recommended Operating Conditions of LQ100X/LQ100 Package in GW1N-2 (LV Version)

Name	Description	Min.	Max.
VCC	Core voltage	1.14V	1.26V
VCCIO0, VCCIO2, VCCIO3, VCCIO4, VCCIO5	I/O Bank voltage	1.14V	3.6V
VCCX/VCCIO1	Auxiliary voltage VCCX and VCCIO1 are internally short-circuited.	1.71V	3.6V

Note!

VCCX should be greater than or equal to the Max. VCCIO.

^[1] When MIPI HARD Core is not used, users can either leave VCCD and VCCIOD floating or connect them to 1.2V power.

^[2] The package is in the EOL stage.

Recommended Operating Conditions of LQ100X/LQ100 Package in GW1N-2 (UV Version)

Name	Description	Min.	Max.
VCCX/VCC	Core voltage VCC and auxiliary voltage VCCX are internally short-circuited.	1.71V	3.6V
VCCIO0, VCCIO1, VCCIO2, VCCIO3, VCCIO4, VCCIO5	I/O Bank voltage	1.14V	3.6V

Recommended Operating Conditions of LQ144X/LQ144 Package in GW1N-2 (LV Version)

Name	Description	Min.	Max.
VCC	Core voltage	1.14V	1.26V
VCCIO0, VCCIO2, VCCIO3, VCCIO4, VCCIO5	I/O Bank voltage	1.14V	3.6V
VCCX/VCCIO1	Auxiliary voltage VCCX and VCCIO1 are internally short-circuited.	1.71V	3.6V

Recommended Operating Conditions of LQ144X/LQ144 Package in GW1N-2 (UV Version)

Name	Description	Min.	Max.
VCCX/VCC	Core voltage VCC and auxiliary voltage VCCX are internally short-circuited.	1.71V	3.6V
VCCIO0, VCCIO1, VCCIO2, VCCIO3, VCCIO4, VCCIO5	I/O Bank voltage	1.14V	3.6V

Recommended Operating Conditions of QN48 Package in GW1N-2

Name	Description	Min.	Max.
VCC	Core voltage	LV version	1.14V
		UV version	1.71V
VCCIO0, VCCIO1	I/O Bank voltage	1.14V	3.6V
VCCIO3/VCCIO4/VCCIO5	I/O Bank voltage, VCCIO3, VCCIO4, and VCCIO5 are internally short-circuited.	1.14V	3.6V
VCCX/VCCIO2	Auxiliary voltage VCCX and VCCIO2 are internally short-circuited.	1.71V	3.6V

Note !

It is highly recommended that the EPAD connect to GND, but not a requirement.

Note!

VCCX should be greater than or equal to the Max. VCCIO.

^[1] When MIPI HARD Core is not used, users can either leave VCCD and VCCIOD floating or connect them to 1.2V power.

^[2] The package is in the EOL stage.

Recommended Operating Conditions of QN48H Package in GW1N-2

Name	Description	Min.	Max.
VCC/VCCIO1	Core voltage VCC and I/O Bank voltage VCCIO1 are internally short-circuited.	1.14V	1.26V
VCCIO0	I/O Bank voltage	1.14V	3.6V
VCCIO3/VCCIO4/VCCIO5	I/O Bank voltage, VCCIO3, VCCIO4, and VCCIO5 are internally short-circuited.	1.14V	3.6V
VCCX/VCCIO2	Auxiliary voltage VCCX and VCCIO2 are internally short-circuited.	1.71V	3.6V
VCCD/VCCIOD ^[1]	VCCD and VCCIOD are internally short-circuited.	1.14V	1.26V

Note !

It is highly recommended that the EPAD connect to GND, but not a requirement.

Recommended Operating Conditions of MG132H Package in GW1N-2

Name	Description	Min.	Max.
VCC	Core voltage	1.14V	1.26V
VCCIO0, VCCIO1, VCCIO2, VCCIO3, VCCIO4, VCCIO5	I/O Bank voltage	1.14V	3.6V
VCCD/VCCIOD ^[1]	VCCD and VCCIOD are internally short-circuited.	1.14V	1.26V
VCCX	Auxiliary voltage	1.71V	3.6V

Recommended Operating Conditions of MG49 Package in GW1N-2

Name	Description	Min.	Max.
VCC	Core voltage	1.14V	1.26V
	LV version	1.71V	3.6V
VCCIO0, VCCIO1	I/O Bank voltage	1.14V	3.6V
VCCIO2/VCCIO3/VCCIO4/VCCIO5	I/O Bank voltage, VCCIO2, VCCIO3, VCCIO4, and VCCIO5 are internally short-	1.14V	3.6V
VCCX	Auxiliary voltage	1.71V	3.6V

Note!

VCCX should be greater than or equal to the Max. VCCIO.

^[1] When MIPI HARD Core is not used, users can either leave VCCD and VCCIOD floating or connect them to 1.2V power.

^[2] The package is in the EOL stage.

Recommended Operating Conditions of QN32X Package in GW1N-2 (LV Version)

Name	Description	Min.	Max.
VCCIO1, VCCIO2, VCCIO3	I/O Bank voltage	1.14V	3.6V
VCC/VCCIO4/VCCIO5	Core voltage, I/O Bank voltage VCCIO4 and VCCIO5 are internally short-circuited.	1.14V	1.26V
VCCIO0/VCCX	Auxiliary voltage VCCX and VCCIO0 are internally short-circuited.	1.71V	3.6V

Note !

It is highly recommended that the EPAD connect to GND, but not a requirement.

Recommended Operating Conditions of QN32X Package in GW1N-2 (UV Version)

Name	Description	Min.	Max.
VCCX	Auxiliary voltage VCCX	1.71V	3.6V
VCCIO0, VCCIO1, VCCIO2, VCCIO3	I/O Bank voltage	1.14V	3.6V
VCC/VCCIO4/VCCIO5/VCCX	Core voltage, auxiliary voltage VCCX, I/O Bank voltage VCCIO4 and VCCIO5 are internally short-circuited.	1.71V	3.6V

Note !

It is highly recommended that the EPAD connect to GND, but not a requirement.

Recommended Operating Conditions of QN88 Package in GW1N-2

Name	Description	Min.	Max.
VCC	Core voltage	LV version	1.14V
		UV version	1.71V
VCCIO0, VCCIO1, VCCIO2, VCCIO3, VCCIO4, VCCIO5	I/O Bank voltage	1.14V	3.6V
VCCD/VCCIOD ^[1]	VCCD and VCCIOD are internally short-circuited.	1.14V	1.26V
VCCX	Auxiliary voltage	1.71V	3.6V

Note !

It is highly recommended that the EPAD connect to GND, but not a requirement.

Note!

VCCX should be greater than or equal to the Max. VCCIO.

^[1] When MIPI HARD Core is not used, users can either leave VCCD and VCCIOD floating or connect them to 1.2V power.

^[2] The package is in the EOL stage.

Recommended Operating Conditions of CS42H Package in GW1N-2

Name	Description	Min.	Max.
VCC	Core voltage	1.14V	1.26V
VCCIO0, VCCIO1, VCCIO2	I/O Bank voltage	1.14V	3.6V
VCCIO3/VCCIO4/VCCIO5	I/O Bank voltage, VCCIO3, VCCIO4, and VCCIO5 are internally short-circuited.	1.14V	3.6V
VCCD/VCCIOD ^[1]	VCCD and VCCIOD are internally short-circuited.	1.14V	1.26V
VCCX	Auxiliary voltage	1.71V	3.6V

Recommended Operating Conditions of LQ144F^[2] Package in GW1N-2

Name	Description	Min.	Max.
VCC	Core voltage	1.14V	1.26V
VCCIO0, VCCIO2, VCCIO3, VCCIO4, VCCIO5	I/O Bank voltage	1.14V	3.6V
VCCIO1/VCCX	Auxiliary voltage VCCX and VCCIO are internally short-circuited.	1.71V	3.6V

Recommended Operating Conditions of QN32 Package in GW1N-2 (LV Version)

Name	Description	Min.	Max.
VCCIO1, VCCIO2, VCCIO3	I/O Bank voltage	1.14V	3.6V
VCC/VCCIO4/VCCIO5	Core voltage, I/O Bank voltage VCCIO4 and VCCIO5 are internally short-circuited.	1.14V	1.26V
VCCIO0/VCCX	Auxiliary voltage VCCX and VCCIO0 are internally short-circuited.	1.71V	3.6V

Note !

It is highly recommended that the EPAD connect to GND, but not a requirement.

Recommended Operating Conditions of QN32 Package in GW1N-2 (UV Version)

Name	Description	Min.	Max.
VCCIO0, VCCIO1, VCCIO2, VCCIO3	I/O Bank voltage	1.14V	3.6V
VCC/VCCIO4/VCCIO5/VCCX	Core voltage, auxiliary voltage VCCX, I/O Bank voltage VCCIO4 and VCCIO5 are internally short-circuited.	1.71V	3.6V
VCCX	Auxiliary voltage	1.71V	3.6V

Note !

It is highly recommended that the EPAD connect to GND, but not a requirement.

Note!

VCCX should be greater than or equal to the Max. VCCIO.

^[1] When MIPI HARD Core is not used, users can either leave VCCD and VCCIOD floating or connect them to 1.2V power.

^[2] The package is in the EOL stage.

Recommended Operating Conditions of CS100H Package in GW1N-2

Name	Description	Min.	Max.
VCC	Core voltage	1.14V	1.26V
VCCIO0, VCCIO1, VCCIO2, VCCIO3, VCCIO4, VCCIO5	I/O Bank voltage	1.14V	3.6V
VCCD/VCCIOD ^[1]	VCCD and VCCIOD are internally short-circuited.	1.14V	1.26V
VCCX	Auxiliary voltage	1.71V	3.6V

Recommended Operating Conditions of LQ144F Package in GW1N-2

Name	Description	Min.	Max.
VCC	Core voltage	1.14V	1.26V
VCCIO0, VCCIO2, VCCIO3, VCCIO4, VCCIO5	I/O Bank voltage	1.14V	3.6V
VCCIO1/VCCX	Auxiliary voltage VCCX and VCCIO are internally short-circuited.	1.71V	3.6V